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Preparation and properties of autoclaved aerated concrete using coal gangue and iron ore tailings

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HIGHLIGHTS

- The optimum activation temperature for coal gangue found in this study is 600 °C.
- A new type of autoclaved aerated concrete is developed using the coal gangue and iron ore tailings.

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ABSTRACT

In this paper a new type of autoclaved aerated concrete (ACC) was developed using coal gangue (CGC) and iron ore tailings (ITOs). The material compositions, calcination temperatures of the coal gangue, and the composition of the hydration products were analysed using differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), X-ray diffraction (XRD), and scanning electron microscope (SEM). The optimal thermal activation temperatures and the activation mechanisms of the calcined coal gangue were also studied. The results indicated that the optimal calcination temperature of coal gangue was approximately 600 °C. The bulk density and compressive strength of AAC samples were approximately 609 kg m^{-3} and 3.68 MPa, respectively, which passed the requirements of A3.5, B06 level of AAC sample regulated by GB/T 11969-2008. Before the autoclaving, the hydration products in ACC containing CGC and ITOs were tobermorite, hibschite, ettringite, and C–S–H gel; during the autoclaving, ettringite was decomposed and more tobermorite formed coupled with C–S–H gel and hibschite.
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1. Introduction

Coal gangue is one type of industrial solid waste that is discharged when raw coal is excavated and washed after mining. Based on statistics [1], every ton of coal production will cause 0.15–0.2 metric tons of coal gangue to be generated. In 2012, the total of coal gangue stockpiled in China reached 5 billion metric tons and was increasing at a rate of 0.15–0.2 billion metric tons per year [2,3]. Coal gangue has become one of the major industrial solid wastes in China. The primary disposal method of the coal gangue waste is storage, which can easily cause the coal gangue to weather and spontaneously combust during long periods of exposure, resulting in environmental problems, such as dust generation, gas emissions (CO, CO_2, SO_2, H_2S, NO_X) and heavy metal contamination [4]. However, coal gangue may constitute a useful by-product that can be used to generate electricity [4,5], used to produce cement materials [6–8] and glass–ceramics [9] and used to backfill mines [10–12], although the rate of reuse is lower than 15%. The primary chemical constituents of coal gangue are SiO_2, Al_2O_3 and similar mineral oxides, and its primary mineralogical phases are feldspar, quartz and similar silicates [13]. Therefore, reactions between coal gangue and water are very slow [14], and the cementitious ability of the original coal gangue is very low. Kim and Hong reported that calcination was an effective method for stimulating the activity of coal gangue [15]. At high temperatures, the atomic movements in the coal gangue were significantly enhanced, resulting in the rapid removal of bonded water from the minerals and the substitution of magnesium, calcium, iron and other cationic atoms in interstitial positions. Calcination may also

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cause insufficient bonding between the silica and alumina octahedra, thus generating a great deal of bond breakage at the free ends. Research has shown that coal gangue calcined at 700 °C exhibited greater activity [16]. Therefore, it could be expected that the activity of coal gangue would be improved by calcination. After calcination or spontaneous combustion, the broken Al–O and Si–O bonds will improve the chemical activity of the material, resulting in higher cementitious activity.

Iron ore tailings are also an industrial solid waste; these mostly result from the process of iron production. In recent years, the amount of the iron ore tailings has grown rapidly due to rapid development of the steel industry in China. From 2007 to 2011, the stockpiles of iron ore tailings increased by 2.899 billion metric tons; in 2011 in particular, approximately 806 million metric tons were generated. However, the comprehensive recycling of these tailings totalled only 307 million metric tons. To reduce the amount of this waste, a new technology has been applied in recent years in which iron ore tailings are used as the primary siliceous raw material to produce autoclaved aerated concrete (ACC). Mechanical grinding of the iron ore tailings has been used to reduce their grain size and increase their chemical activity when they are used as siliceous materials in AAC [17–19].

In view of the energy savings and waste stream reduction via recycling, a new type of AAC material was developed using waste coal gangue and iron ore tailings materials. In this study, coal gangue first underwent a low-temperature activation treatment and was then crushed and added to the iron ore tailings to produce the AAC. The coal gangue calcination temperature, phase transformation process and activation reaction mechanism were also investigated. Meanwhile, the mechanical and physical properties of the AAC were also analysed in detail. This technology represents a new process for the use of coal gangue and iron ore tailings for the purpose of comprehensive utilisation of solid waste in China.

2. Materials and methods

2.1. Experimental materials

The AAC samples were prepared using the following raw materials: coal gangue, autoclaved coal, iron ore tailings, lime, P·O 42.5 Portland cement and desulphurisation gypsum. The chemical compositions of the raw materials are listed in Table 1.

2.1.1. Portland cement

The initial and final setting times of the P·O 42.5 Portland cement were 118 min and 190 min, respectively.

2.1.2. Iron ore tailings

The iron ore tailings were from Beijing and its mineralogic phases were identified by XRD, as shown in Fig. 1. Table 1 shows that the amount of SiO_2 in the iron ore tailings was approximately 68.96%, and the level of residual iron in the tailing, which was present in the form of magnetite, was approximately 5.2%. The iron ore tailings thus were of a high-silicon low-iron magnetite–quartzite type.

2.1.3. Coal gangue

The coal gangue used in this study was from Heilongjiang. The mineral phases in the coal gangue are shown in Fig. 2, which indicates quartz and clay minerals (kaolinite, montmorillonite, illite, and muscovite) as the main mineral phases, and siderite and anorthite as the minor phases. Quartz and anorthite are highly inert phases as they belong to the trigonal and triclinic crystal systems. In contrast, kaolinite, illite and montmorillonite are layered silicates which are easily altered with increased levels of chemical activity. Hence, it may be inferred that the high activity of the coal gangue was primarily due to the activation of clay minerals at relatively low temperatures.

2.1.4. Lime

The digestion time of the lime was 13 min, and the digestion temperature was 66 °C.

2.1.5. Desulphurisation gypsum

The primary mineral phase of desulphurisation gypsum is $CaSO_4{\cdot}2H_2O$.

2.1.6. Aluminium powder

Aluminium powder was used as a gas-producing agent for foaming of the slurry.

2.2. Material preparation

First, the coal gangue was crushed to a particle size of less than 2 mm. Then, the powder was dried and further crushed to a fine powder with a specific surface area of 321 $m^2\ kg^{-1}$ using a ball mill. Finally, the coal gangue samples were calcined in a CD-1400X muffle furnace for 4 h at five temperatures (500 °C, 600 °C, 700 °C, 800 °C, 900 °C) and then water quenched. The iron ore tailings were first dried and then crushed using a ball mill to yield a specific surface area of 317 $m^2\ kg^{-1}$.

2.2.1. Preparation of AAC

The mass proportions of CGC in the dry mixtures were 40%, 38%, 35%, 33%, 30%, 28%, 25%, 23%, 20%, 18%, 15%, 13%, 10%, 8%, 5%, 3% and 1%, respectively. The corresponding mass proportions of crushed iron ore tailings were 20%, 22%, 25%, 27%, 30%, 32%, 35%, 37%, 40%, 42%, 45%, 47%, 50%, 52%, 55%, 57% and 59%, respectively. The amounts of lime, cement and gypsum in each dry mixtures were 25%, 10%, and 5%, respectively, and those of aluminium powder and water were 0.06% and 60% by mass of dry mixtures. The AAC samples with these various amounts of CGC were denoted as CT1–CT17.

Warm water (55 ± 1 °C) was added into the well mixed raw material powders and the slurries were stirred for 90 s. Then aluminium powder was added into the slurries with additional 40 s stirring. The obtained slurries were cast into preheated steel moulds with dimensions of 100 mm × 100 mm × 100 mm, and they were allowed to expand and harden at 55 ± 1 °C for 3 h in a chamber. After removing the expanded parts above the top surface of the moulds, the trimmed samples were de-molded and placed in an industrial autoclave for 8 h at 185 °C with a pressure of 12.5 bars.

2.3. Sample characterisation

The bulk density and compressive strength of the AAC samples were tested according to GB/T 11968-2008. For bulk density measurements, three samples of each AAC mixture were oven dried at 60 ± 5 °C for 24 h, then at 80 ± 5 °C for 24 h, and finally at 105 ± 5 °C for 24 h, consecutively. The mass of the oven dried samples were used to calculate the bulk density. The compressive strength were performed on three samples of each AAC mixture with a moisture content of 8–12% at a loading rate of 2.0 ± 0.5 kN/s.

The X-ray diffraction (XRD) spectra of the coal gangue, iron ore tailings and AAC samples were performed using a D/Max-RC diffractometer (Japan) with Cu Kα radiation, voltage of 40 kV, current of 150 mA and 2θ scanning ranging between 5° and 90°. The DSC–TG analysis of the coal gangue was performed from 20 to 1000 °C at a rate of 10 °C/min using a Netzsch STA 449C thermal analyser with dry air as the stripping gas. FESEM observation was performed to analyse the hydration products of the AAC samples using a Zeiss SUPRA™55 scanning electron microscope coupled with a Be4-U92 energy spectrum.

Table 1
Chemical composition of raw materials (wt.%).

Material	SiO_2	Al_2O_3	Fe_2O_3	FeO	MgO	CaO	Na_2O	K_2O	SO_2	Loss
Coal gangue	34.05	26.00	0.49	1.70	0.61	0.67	0.00	0.16	0.28	32.76
Autoclaved coal	51.05	36.71	3.12	0.42	0.92	1.05	0.00	0.23	0	0.72
Iron ore tailings	68.96	7.68	2.32	4.47	3.64	4.35	1.41	1.85	0.024	2.49
Lime	5.45	3.85	1.68	0.08	3.56	78.76	–	1.25	0.45	3.93
Cement	25.06	6.10	3.31	0.21	3.87	55.56	0.23	0.95	–	4.16
Desulphurisation gypsum	2.84	0.78	0.25	0.03	0.47	40.13	0.14	0.12	33.21	–

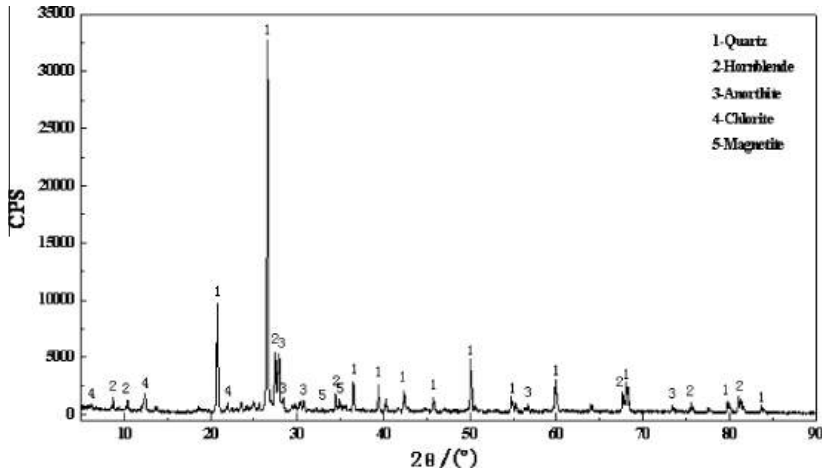

Fig. 1. Mineral phases in iron ore tailings.

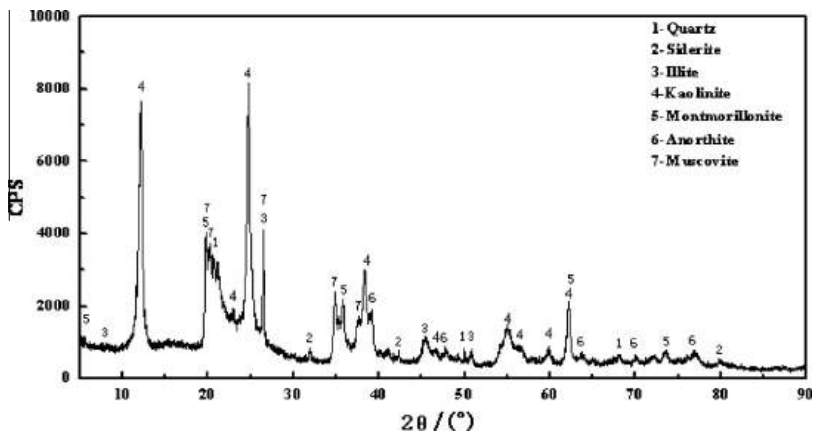

Fig. 2. Mineral phases in coal gangue.

3. Results and discussion

3.1. Effect of calcination temperature on the activity of coal gangue

Details of the relationships between the temperatures and physical properties of the coal gangue are illustrated in Fig. 3.

The result of the DSC–TG analysis of the coal gangue is shown in Fig. 3, which indicates that the coal gangue continuously lost weight during the heating process. The weight loss of the coal gangue occurred at temperatures from 250 °C to approximately 600 °C. A distinct endothermic peak can be observed on the DSC curve at 379 °C, which may be the result of the loss of free water and adsorbed water below 379 °C. An exothermic peak has also been observed between 250 and 600 °C. The cumulative weight loss was 25.21% as the temperature increased. A valley of heat release appeared at 451 °C due to combustion of residual coal in the coal gangue and decomposition and oxidisation of siderite at 400–500 °C [20]. Later, there was an endothermic peak at 525 °C, which corresponds to the removal of hydroxyl in kaolinite and the formation of metakaolinite [21]. There was a 0.99% weight loss in the sample in the temperature range of 600–1000 °C, which may be due to the dehydroxylation of white mica at 850–950 °C and the phase transformation of the metakaolinite into amorphous $A1_2O_3$ and SiO_2 phases at 976 °C [21]. The industrial analytical results of the coal gangue were 0.43% air dry basis moisture, 72.81% dry ash, 14.98% dry volatiles and 10.78% fixed carbon, based on Chinese standard GB/T 212-2008, which were essentially equivalent to the results from the DSC–TG analyses.

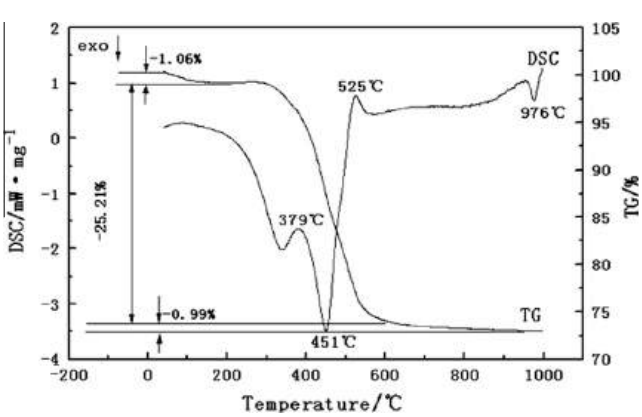

Fig. 3. DSC–TG curves of coal gangue.

The chemical activity of the coal gangue could primarily result from the thermal decomposition and glass transformation reactions of the clay minerals (e.g., $Al_2O_3{\cdot}2SiO_2{\cdot}2H_2O$). During heating, a series of phase transformations of $Al_2O_3{\cdot}2SiO_2{\cdot}2H_2O$ occurred as a result of various microstructures and atomic activities. No molecular water or free atoms were present in $Al_2O_3{\cdot}2SiO_2{\cdot}2H_2O$, which consists of layered silica tetrahedra and alumina octahedra bonded with hydrogen bonds. However, when the coal gangue was heated

to 500–700 °C, these bonds were broken, and it leaded to the formation of an amorphous phase $Al_2O_3{\cdot}2SiO_2$ [22,23]. As of the disordered atomic arrangement in amorphous $Al_2O_3{\cdot}2SiO_2$, the thermal process contributed to the activation of pozzolanic minerals containing large amounts of Si and Al.

The thermal activation of the gangue at certain calcination temperatures in the range of 700–900 °C may be considered as dehydration of $Al_2O_3{\cdot}2SiO_2{\cdot}2H_2O$, which comes with the transformation of ordered $Al_2O_3{\cdot}2SiO_2{\cdot}2H_2O$ structure to amorphous, disordered $Al_2O_3{\cdot}2SiO_2$, SiO_2 and Al_2O_3 structures [24]:

$$\underset{\text{(kaolinite)}}{Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O} \rightarrow \underset{\text{(metakaolinite)}}{Al_2O_3 \cdot 2SiO_2} + 2H_2O \quad (1)$$

at 700–900 °C,

$$\underset{\text{(metakaolinite)}}{Al_2O_3 \cdot 2SiO_2} \rightarrow \underset{\text{(metastable } SiO_2\text{)}}{Al_2O_3 + 2SiO_2} \quad (2)$$

When the temperature was higher than 1000 °C, the products of decomposition, SiO_2 and Al_2O_3, reacted further to form mullite:

$$\underset{\text{(metastable } SiO_2\text{)}}{2SiO_2} + 3Al_2O_3 \rightarrow \underset{\text{(mullite)}}{3Al_2O_3 \cdot 2SiO_2} \quad (3)$$

However, if the calcination temperature was too low, the included carbon burned out insufficiently, resulting in a decreased level of activity.

Based on the results shown in Figs. 2 and 3, the cement was blended with the coal gangue calcined at five various temperatures (500 °C, 600 °C, 700 °C, 800 °C, 900 °C) at a replacement rate of 30%. Details of the sample compositions and their mechanical properties are shown in Table 2.

Table 2 shows that the mechanical strength of the mortars with coal gangue (CGC) was lower than that of the ordinary Portland cement mortar: samples CGC1 and CGC5 exhibited compressive strengths of 34.83 MPa and 39.78 MPa, respectively (<42.5 MPa), after 28 days of curing time. The strength of all the other mortar samples exceeded the 42.5-grade cement standard; in particular, the CGC2 samples displayed compressive strengths of 45.82 MPa. Thus, the optimal calcination was approximately 600 °C; there was a small amount carbon of approximately 0.63%.

Table 2
Mechanical properties of hydrous block.

Sample ID	Flexural strength (MPa)		Compressive strength (MPa)	
	3 days	28 days	3 days	28 days
CGC0	7.52	10.38	39.15	59.99
CGC1	4.92	8.17	18.35	34.83
CGC2	6.53	10.62	24.83	45.82
CGC3	6.09	9.46	24.15	44.11
CGC4	5.67	9.55	25.16	42.57
CGC5	5.91	8.93	21.35	39.78

The mineral phases in the CGC calcined at 600 °C are shown in Fig. 4, which evidently indicates that the peaks belonging to kaolinite, illite and montmorillonite are no longer presented, and the remaining phases are anorthite, quartz and muscovite. During calcination, the siderite was decomposed and oxidised to hematite and a small amount of magnetite. After calcination, the much sharper peaks of the quartz suggests its improved crystallisation. Meanwhile, the broadened "convex closure" between 20° and 30° confirms the decomposition of the clay minerals illite and montmorillonite and the formation of amorphous metakaolinite [25].

3.2. Effect of coal gangue dosage on the properties of AAC

Fig. 5 shows that the bulk density and compressive strength of the varying AAC samples. The bulk density of the AAC samples increased from 581 kg m^{-3} to 621 kg m^{-3} as the CGC dosage decreased from 40% to 1%. The addition of CGC thus had a negative effect on the bulk density of the AAC. In sample CT1 containing 40% CGC the iron ore tailing particles were not able to efficiently fill the space between the hydration products. In contrast, in sample CT17 containing 1% CGC and 59% iron ore tailings, there were sufficient particles of iron ore tailings to fill the space so that CT17 had the highest bulk density.

As the CGC dosage decreased, the compressive strength of the samples increased from 2.10 MPa to 4.56 MPa. The compressive strengths of samples from CT9 to CT17 increased from 3.68 to 4.56 MPa, and these samples were qualified with the specification (GB/T 11968-2006) of the A3.5, B06 grade. In addition, the slurries of CT1 and CT2 had high viscosity, high gelation speeds and non-uniform gas porosity distributions. Additionally, there were cracks and defects perpendicular to their surfaces. The observed cracks suggest that the CGC dosage in AAC should be carefully controlled to achieve the best product quality, and the ACC samples with a 20% CGC dosage exhibited the best performance in this study.

3.3. The composition and morphology of AAC

Before and after the autoclaving process, XRD was performed on CT9 to verify the phase transformation. Before being autoclaved,

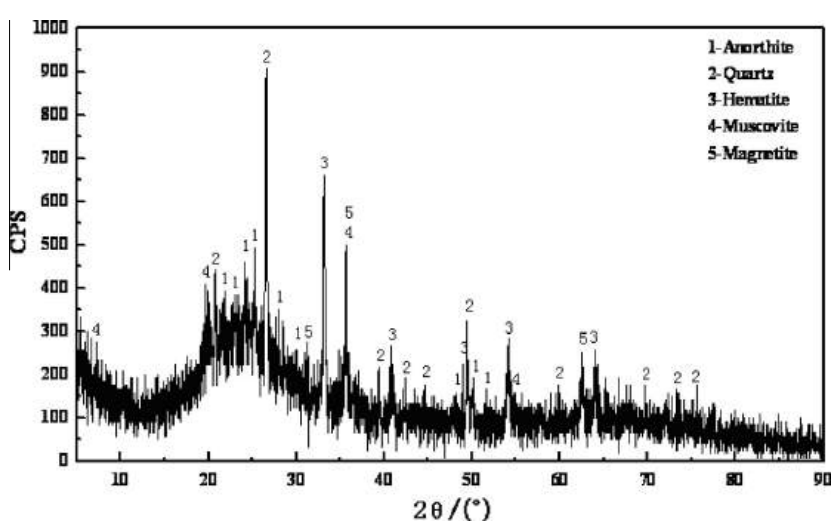

Fig. 4. Mineral phases in coal gangue calcined at 600 °C.

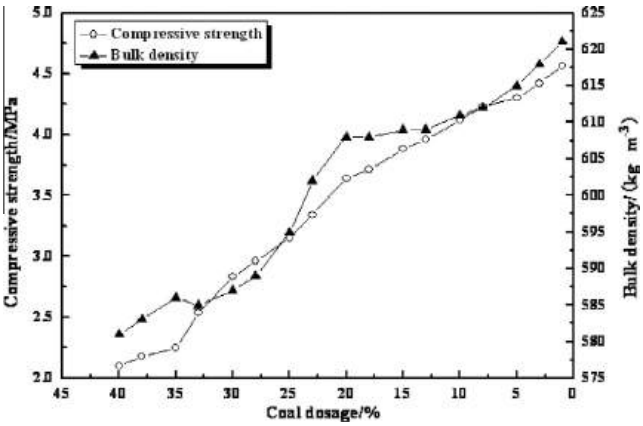

Fig. 5. The effects of various amounts of calcined coal gangue on sample properties.

the primary phases in CT9 were 1.1-nm tobermorite, hibschite, ettringite, portlandite, calcite, quartz, anorthite, hornblende, anhydrite, chlorite, muscovite, magnetite and hematite. After being autoclaved, the primary phases in CT9 were 1.1-nm tobermorite, hibschite, calcite, ferrotschermakite, quartz, anhydrite, chlorite, muscovite, magnetite and hematite.

These findings indicate that before the autoclaving, an excessive amount of $Ca(OH)_2$ was present with insufficient amount of dissolved Si^{4+} ions from CGC and iron ore tailings, resulting in the formation of a small amount of tobermorite, as shown in Fig. 6. After the autoclaving, a large amount of Si^{4+} was dissolved from both CGC and iron ore tailings in alkaline hydrothermal conditions, which promoted the formation of tobermorite. Meanwhile, the reduced intensity of the quartz peaks indicates that more tobermorite formed and $Ca(OH)_2$ was depleted during the autoclaving since the diffraction peaks of tobermorite displayed an increasing trend and the peaks of $Ca(OH)_2$ disappeared. Before the autoclaving, ettringite formed by the reaction of calcium aluminate hydrate (C–A–H) from cement with SO_4^{2-} of gypsum in the early hydration period. Then, ettringite was decomposed into monosulfoaluminate hydrate (AFm), Al^{3+} and SO_4^{2-} at a temperature of approximately 70 °C [26]. During the autoclaving, AFm was continuously decomposed into tricalcium aluminate hexahydrate (C_3AH_6) and $CaSO_4$. Thus, after the autoclaving, the diffraction peaks of gypsum in Fig. 6 were.

Hibschite belongs to the hydrogarnet class. Because of the presence of Al in CGC and iron ore tailings, the hydrogarnet phase first

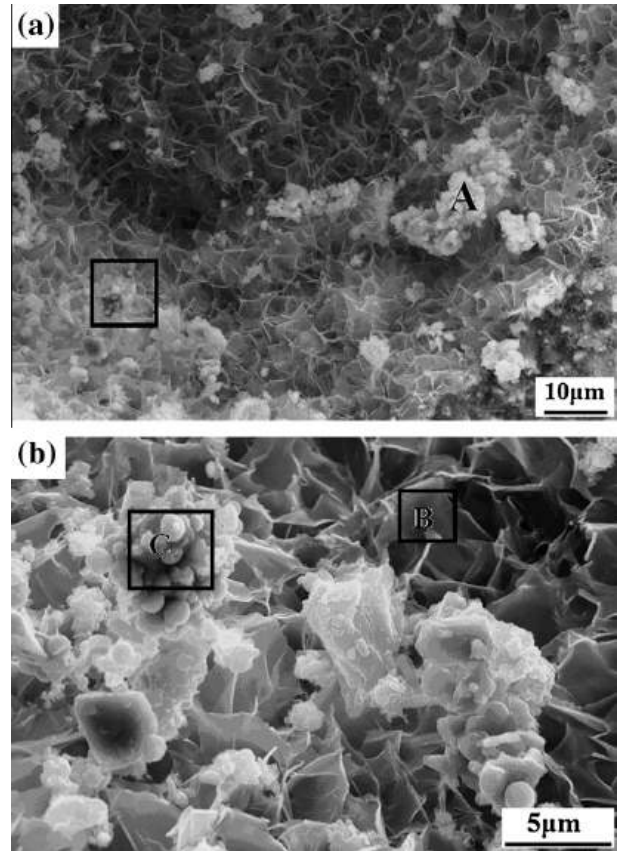

Fig. 7. SEM micrographs of the hydrated products of sample CT9 (a) microstructure of hydrated products of sample CT9; (b) high magnification view of the marked region in (a).

precipitated during the autoclaving. However, continuous autoclaving would help reduce the amount of $[OH^-]$ ions and thus promot another aluminium silicate phase, e.g., tobermorite [27], which would influence the AAC properties [28]. Hibschite was able to precipitate rapidly in a wide temperature range due to massive calcium interaction with the dissolved Si^{4+} and Al^{3+} from both CGC and iron ore tailings during the autoclaving [29]. Because of the limited time of autoclaving, the hydrate hessonite garnet would undergo an incomplete transformation to tobermorite, and thus both were present in the sample, as shown in Fig. 6.

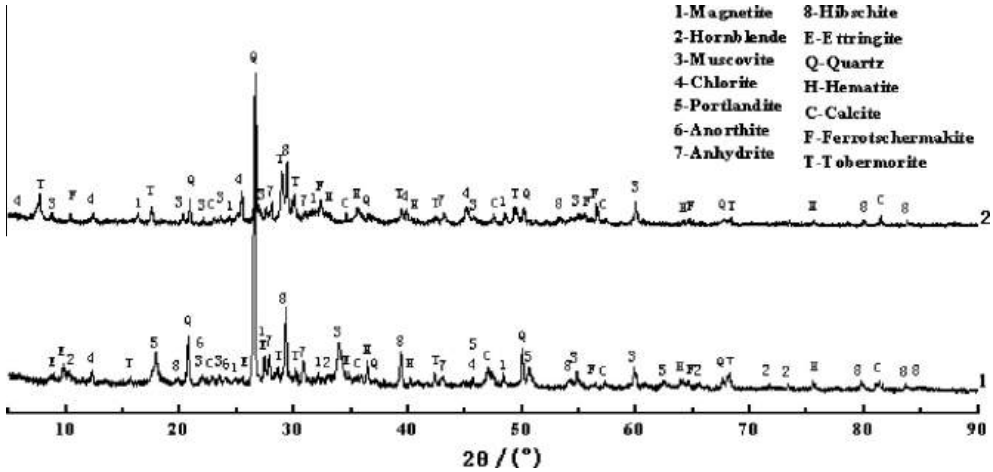

Fig. 6. Mineral phases in AAC samples before and after being autoclaved; 1 – before being autoclaved, 2 – after being autoclaved.

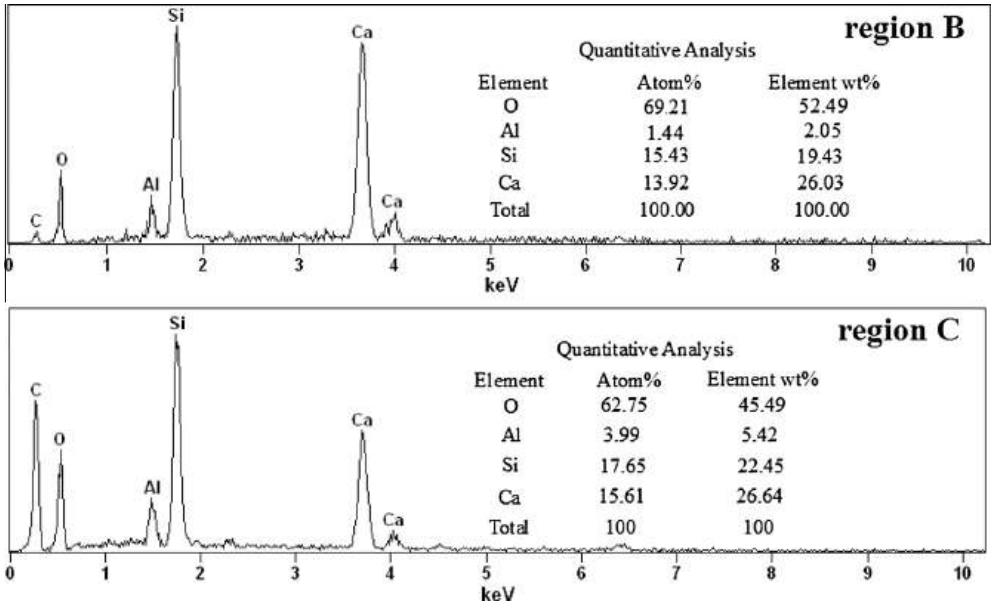

Fig. 8. The EDX spectrum of marked regions B and C in Fig. 7(b).

The ferrotschermakite phase with a double chain structure formed due to ion exchange between $Ca^{2+}/Al^{3+}/Fe^{2+}$ and Si^{4+} during the autoclaving. Anorthite in the raw materials was decomposed and then tobermorite precipitated, which enhanced the material strength. After the autoclaving, the diffraction peak intensity of quartz indicated its reduced amount, which performed as the primary fraction of the aggregate in the samples. $CaCO_3$ also formed because of the carbonation reaction of $Ca(OH)_2$. The $CaCO_3$ together with nonreacted muscovite, chlorite, magnetite and hematite phases would become the primary aggregate in the AAC samples.

As shown in Fig. 6, the broadened "convex closure" in the range of 26–34° indicates that amorphous, weakly crystalline substances in CT9 [25].

The morphologic (SEM) analysis of the hydration products in CT9 are shown in Fig. 7. There were platy tobermorite, weakly crystalline C–S–H gel (at point A in Fig. 7(a)), and hibschite with globular or botryoidalis morphology in the range of 0.5–1 μm (at point B in Fig. 7(a)) in the AAC samples. The C–S–H gel acts as a binder bonding all components in AAC together to form a good network that resists stress concentration under loading [30–32]. This structure may be the reason for the XRD peak broadening in Fig. 6 in the range of 26–34° of 2θ [25].

The EDX results from the selected areas in Fig. 7(b) are shown in Fig. 8. As shown in Fig. 8, the siliceous raw materials contain a large amount of Al ions, which may partially replace Si in the $[SiO_4]$ structure to form $[AlO_4]$. The composition of phase B contains a ratio of nCa/n(Si + Al) = 0.825, which is similar to that of tobermorite (0.833). The composition of phase C contains a ratio of nCa/n(Si + Al) = 0.721, which was equal to that of hibschite (0.721).

4. Conclusions

In this study, the material compositions, calcination temperatures of coal gangue, and physical and mechanical properties and hydration products of AAC were investigated. Based on these analyses, the following conclusions may be drawn:

(1) When the calcination temperatures increased, coal gangue was thermally activated in certain temperature ranges. Based on the strength of the materials, 600 °C was the optimal activation temperature for coal gangue in this study.

(2) A new type of ACC was developed using the coal gangue and the iron ore tailings. The dry mixture proportion of the optimal AAC with a bulk density of 609 kg m^{-3} and compressive strength of 3.68 MPa was 20% CGC, 40% iron ore tailings, 25% lime, 10% cement, 5% desulphurisation gypsum and 0.06% aluminium powder. From this dry mixture, a slurry with 56% water was made.

(3) Before the autoclaving, the hydration products in ACC containing the coal gangue and iron ore tailings were tobermorite, hibschite, ettringite, and C–S–H gel. During the autoclaving, ettringite was decomposed and more tobermorite formed coupled with C–S–H gel and hibschite.

Acknowledgments

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